

MATRIX Final Design Review

MAgnetotail Transition Region In-situ eXplorer



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Magnetotail
Systems
Laboratory

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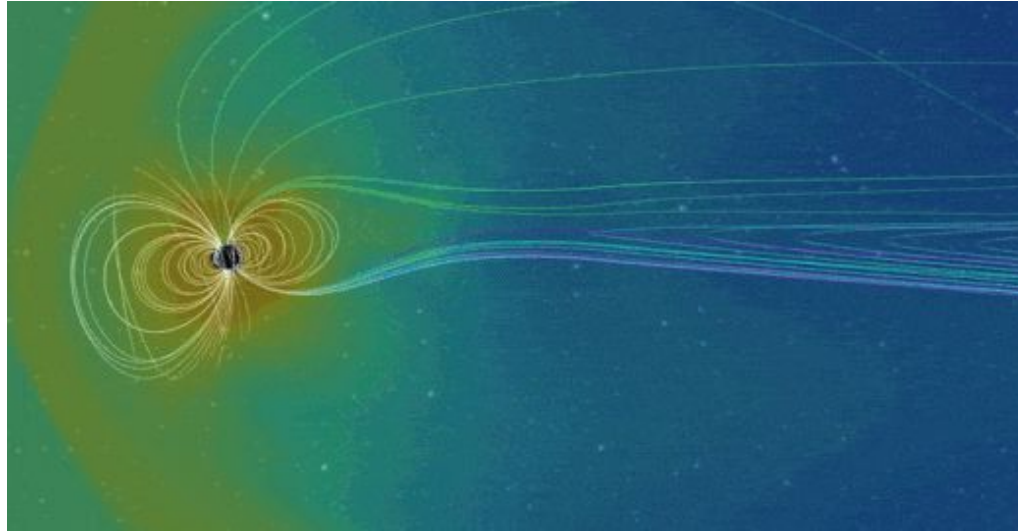
Evan Callas: Energy Storage Lead, POW/PLV

Mission Design

RFP Goals and Objectives: Magnetotail Mesoscale (Topic D)

The **magnetotail** is the night-side of Earth's magnetic field which is blown into a tail-like structure by the solar wind.

Magnetotail transition region: Located $6-12 R_{\text{Earth}}$ “downwind” in the magnetotail and a few R_{Earth} in cross-tail size



Visualization Credit: Tom Bridgman NASA SVS [1]



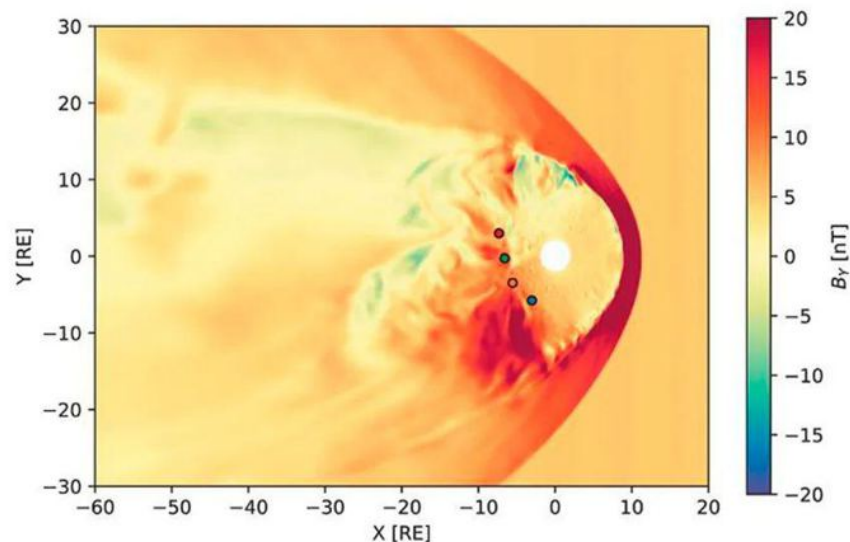
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RFP Goals and Objectives: Magnetotail Mesoscale (Topic D)

- **Why is the MATRIX mission necessary?**
 - Magnetotail transition region: Dynamic, complex, unstudied at the mesoscale
 - Provide **continuous, multi-point, multi-scale, in-situ, field and particle** measurements of the magnetotail transition region
 - Study the evolution and braking of energetic intermittent plasma flows (**Bursty Bulk Flows BBFs**) in the magnetotail
- **Why should I care?**
 - Geomagnetic storms can disrupt GPS and internet satellites, interfere with radio communications on Earth

Simulated distribution of Bursty Bulk Flows



From Merkin et al. (2019) [2]

Mission Objectives & Requirements

Objectives

- Characterize the **spatial and temporal structure of BBFs** across the magnetotail transition region
 - Includes studying the braking of BBFs in the $6.6\text{-}9 R_{\text{Earth}}$ BBF braking zone
- Quantify BBF **magnetic field and particle signatures**
 - To study **energy transport and deposition** in the magnetosphere.

Requirements

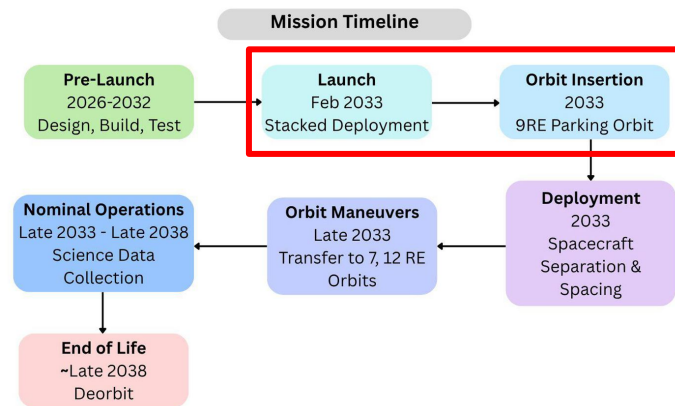
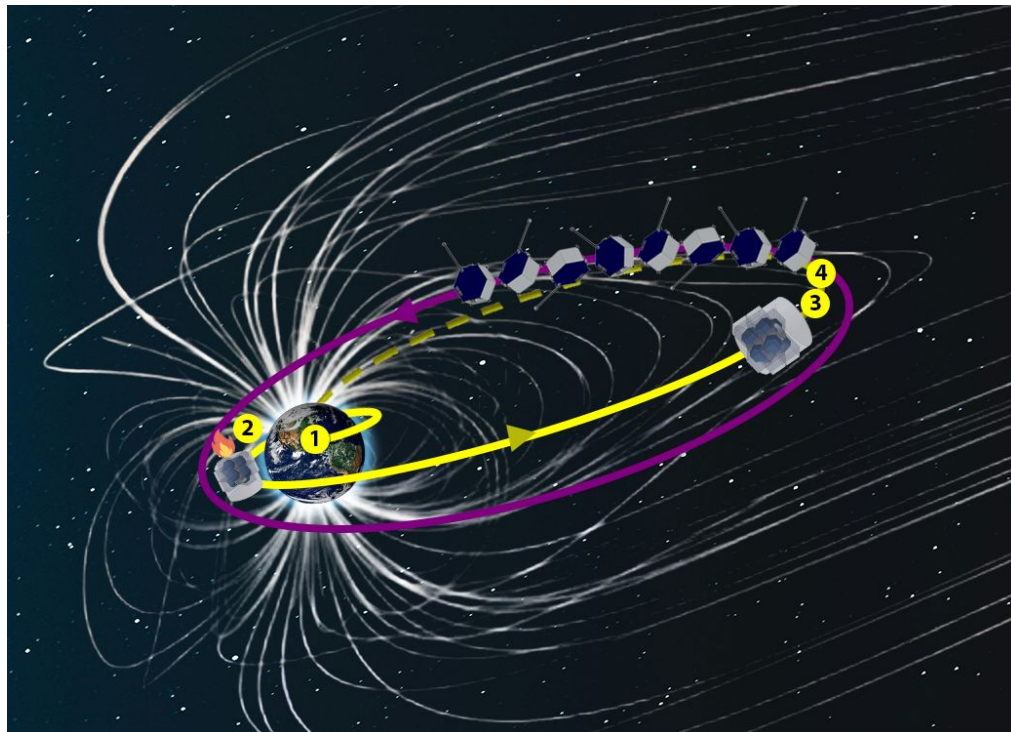
- **Must** obtain **in-situ** magnetic field and plasma measurements in the magnetotail transition region
- **Must** make mesoscale ($1\text{-}3 R_{\text{Earth}}$) spatial measurements of the magnetotail transition region
- The MIDEX-class mission **must** be operational for at least 6 years

Constraints: Cost, Mass, Δv , Power, Pointing

Drivers: Mesoscale Measurements, Science Return, RFP Compliance



Concept of Operations: Launch + Early Orbit



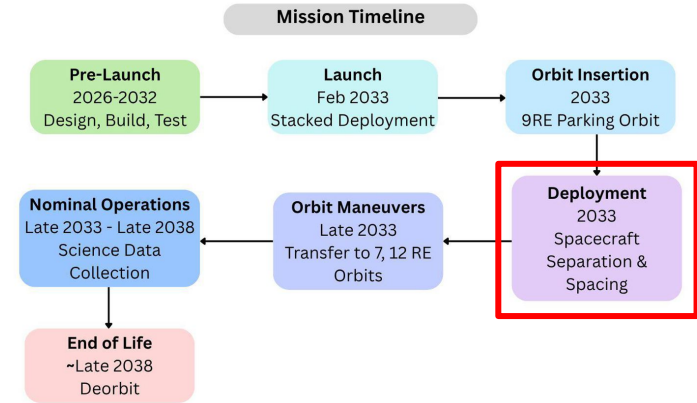
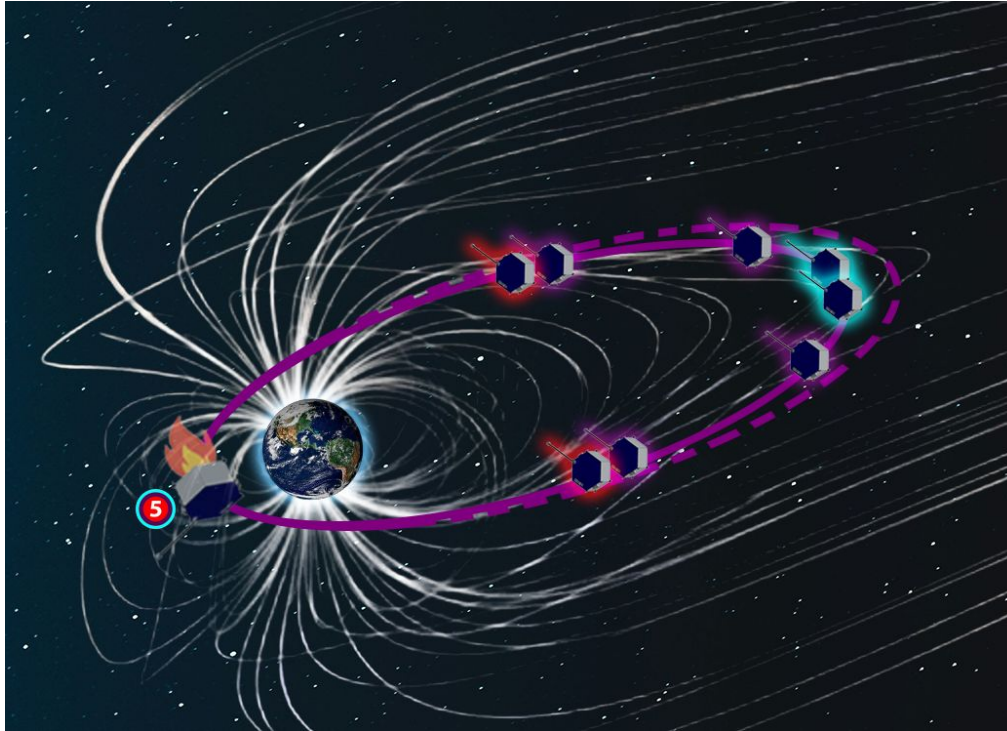
- 1 LAUNCH (FEB 2033)**
Liftoff from Cape Canaveral to Group 2's apogee 9 Earth radii parking orbit
- 2 ORBIT INSERTION BURN (T+HRS)**
- 3 GROUPS 1, 2, 3 PAYLOAD DEPLOYMENT (T+HRS)**
Deployment of all 8 satellites from ESPA rings
- 4 SECOND STAGE DEORBIT BURN (T+HRS)**



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Concept of Operations: Spacing and Commissioning

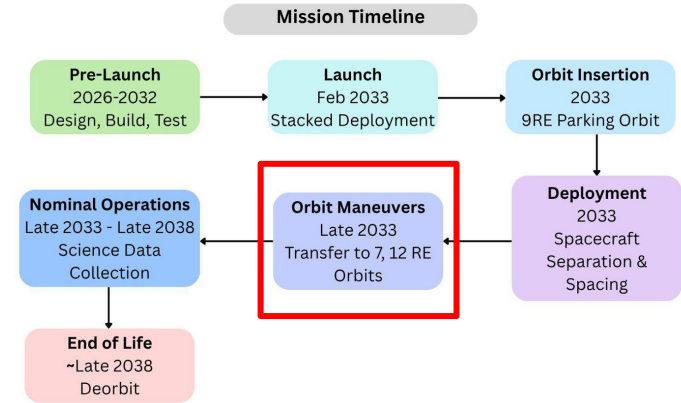
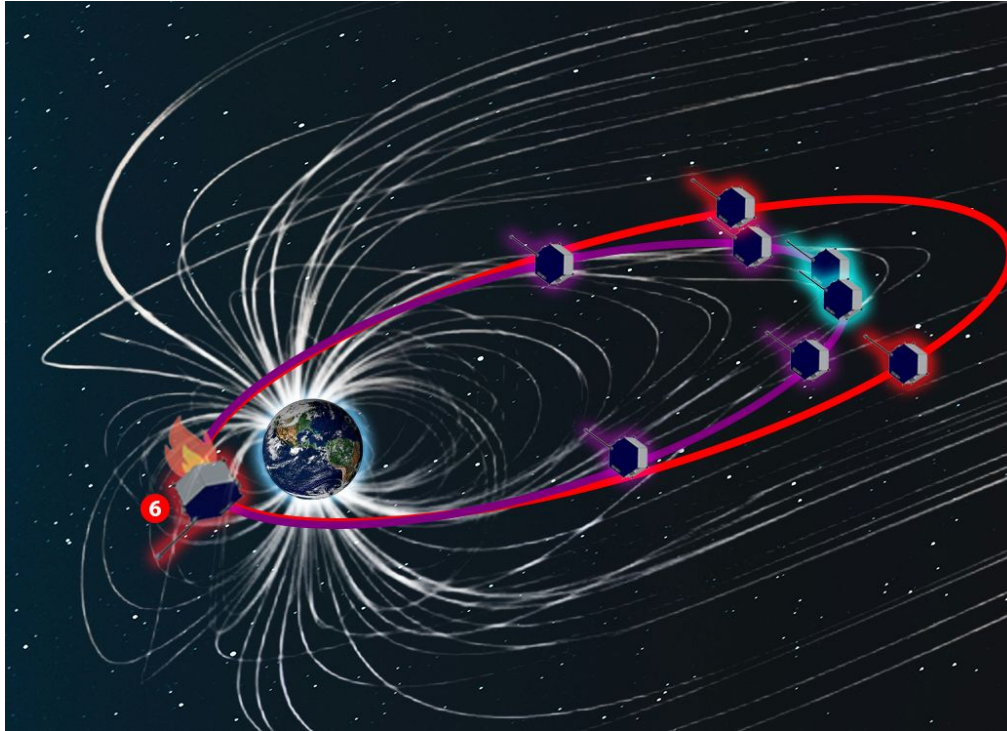


5 GROUPS 1, 2, 3 SPACING AND COMMISSIONING (T+MONTHS)

While in Group 2 orbit, slightly modify orbit apogees so the change in orbital period causes satellites to drift apart

Power on satellites, establish communications, and perform system checkouts

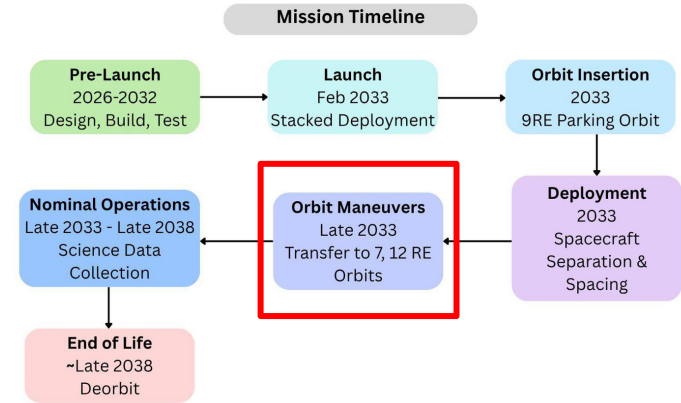
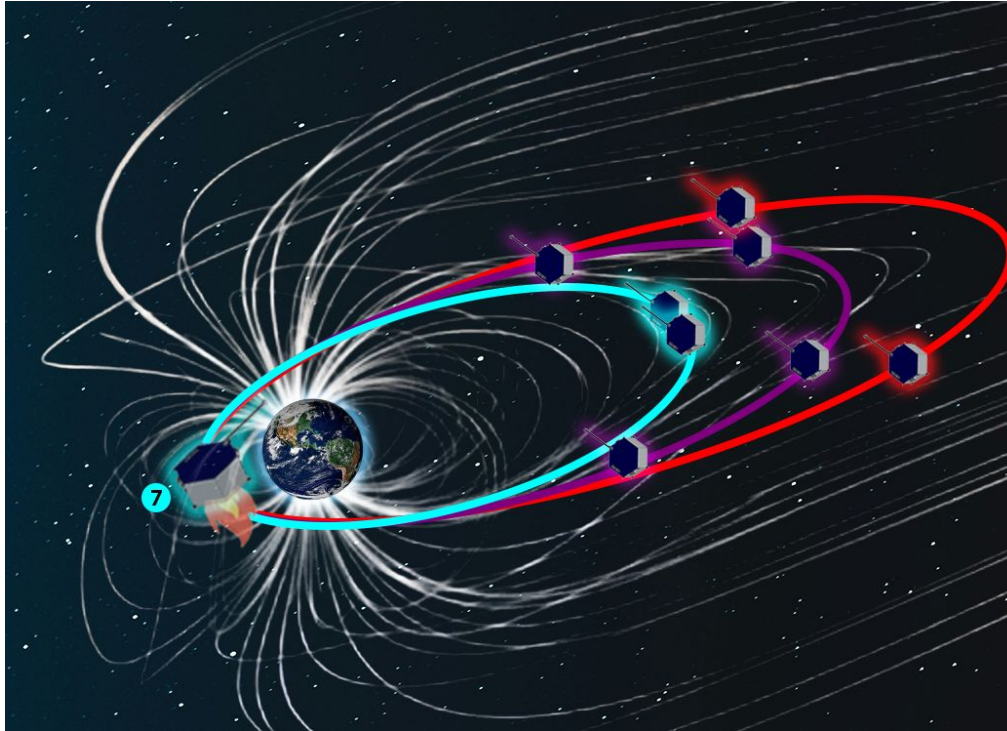
Concept of Operations: Orbit Raising & Lowering



6 GROUP 1 ORBIT RAISING (NOV 2033)

Raise apogee to a radius of 12 Earth radii

Concept of Operations: Orbit Raising & Lowering



7 GROUP 3 ORBIT LOWERING (NOV 2033)

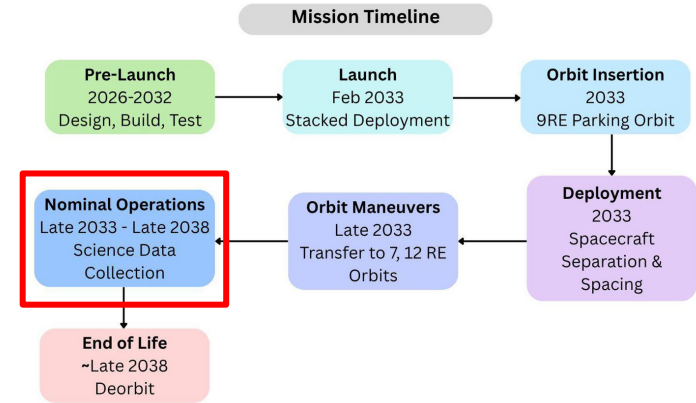
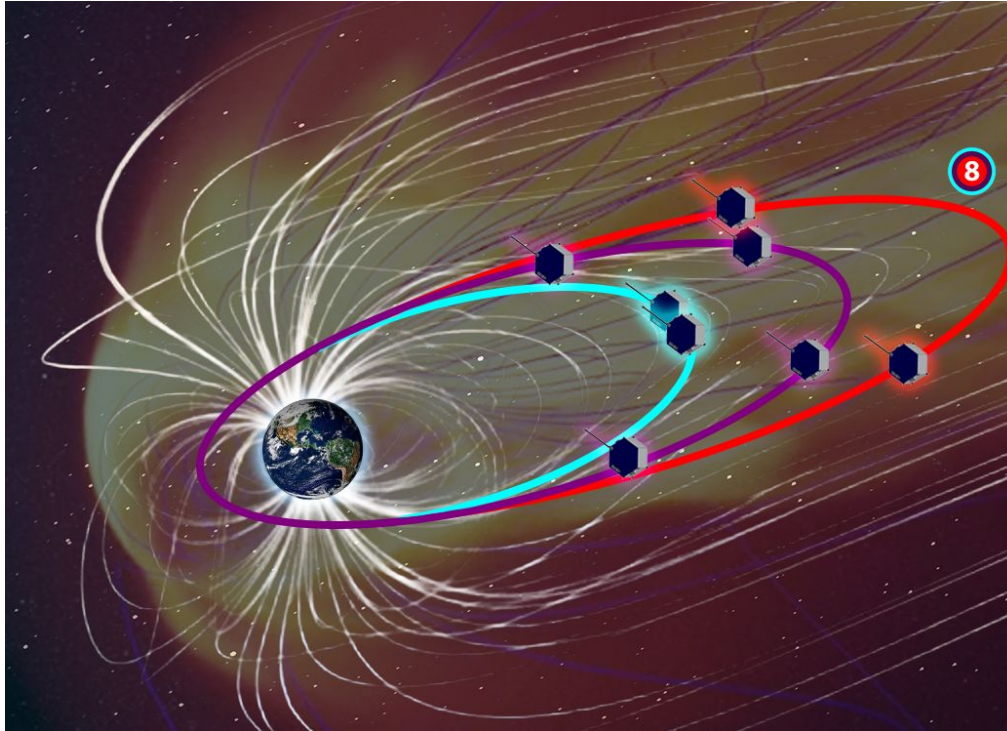
Lower apogee to a radius of 7 Earth radii



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Concept of Operations: Nominal Science Operations



8 5-YEAR PRIME MISSION (STARTS DEC 2033)

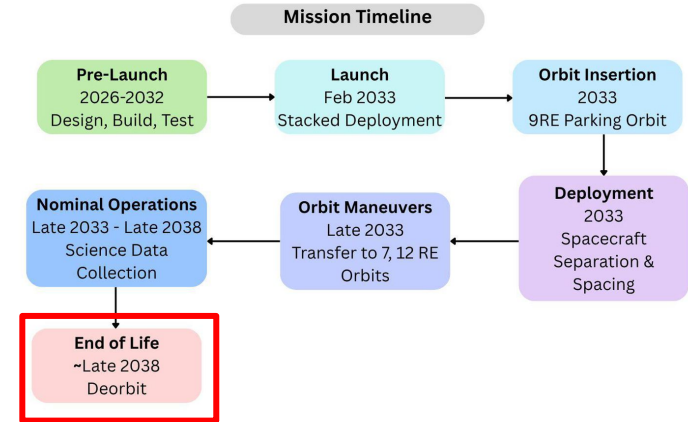
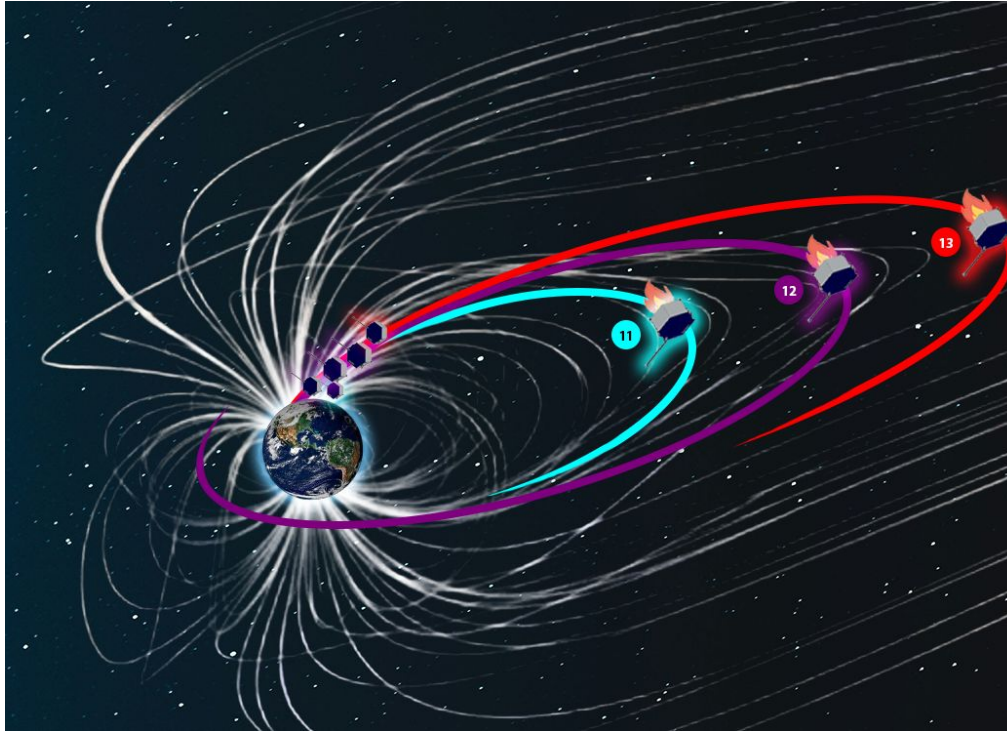
Take in-situ measurements of Earth's magnetotail for approximately half of each year



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Concept of Operations: End-of-Life Deorbit



9 GROUP 1 END-OF-LIFE DEORBIT (>DEC 2038)

10 GROUP 2 END-OF-LIFE DEORBIT (>DEC 2038)

11 GROUP 3 END-OF-LIFE DEORBIT (>DEC 2038)
Disintegrate satellites using atmospheric re-entry



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Spacecraft Modes: Behavior and Payload Configuration

Mode	Spacecraft Behavior	Payload Configuration
Start-Up	System boot, detumble, establish power and comms	Payload OFF
Commissioning	Subsystem checkout and calibration	ESA/EPD OFF; Magnetometer low cadence for calibration
Nominal (Survey)	Routine science ops, stable pointing, periodic downlink	ESA ON, EPD ON; Magnetometer low field range, low cadence
Burst / High-Rate Science	Event-triggered operations, prioritized data storage/downlink	ESA ON, EPD ON; Magnetometer high cadence (up to 128 Hz)
Downlink	Omnidirectional antenna active, data transmission	ESA, EPD OFF; Magnetometer in high field range
Maneuver	Thruster burns, attitude control prioritized	Payload OFF
Eclipse	Power-constrained operations	ESA/EPD ON; Magnetometer switches between low and high cadence depending on Burst or Nominal Survey Modes
Safe Mode	Fault protection, minimal activity, sun-pointing	Payload OFF
Standby / Idle	Awaiting commands, minimal operations	Payload OFF or low-power state
End-of-Life	Deorbit maneuvers and passivation	Payload OFF

Table 1 From the MATRIX OPL PDR Report

Why do we use multiple modes?

→ Modes allow the spacecraft to balance power, data, and safety.

When is science collected?

→ Nominal mode provides continuous survey data.

→ Burst mode captures short, high-value events at high resolution.

What happens under constraints (eclipse/idle)?

→ Power-limited conditions reduce instrument activity.

→ Standby keeps the system in a low-power, command-ready state.

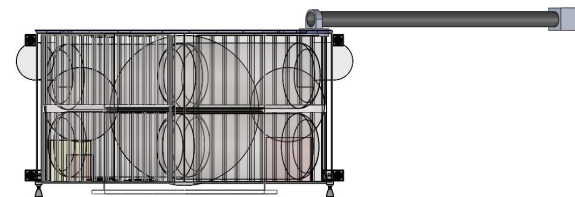
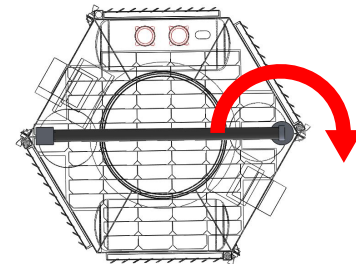
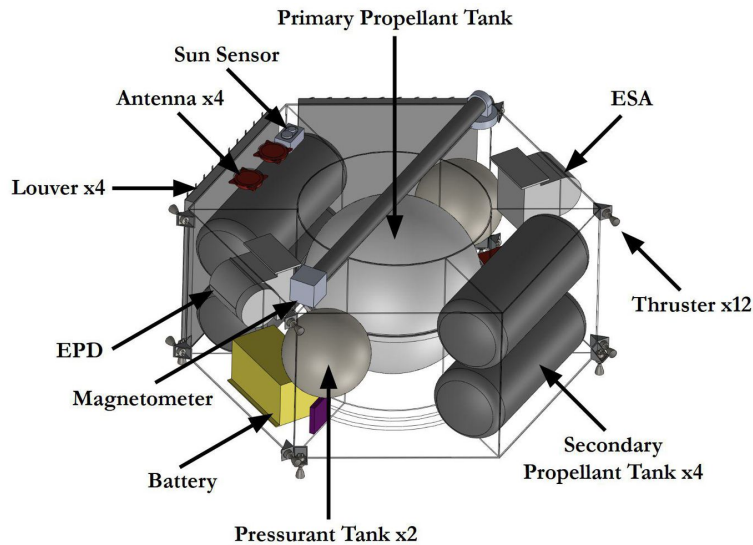
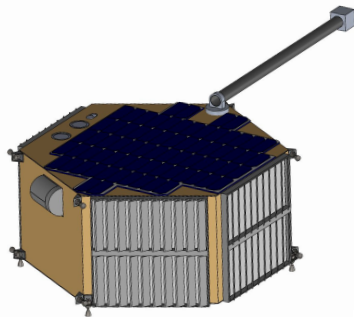
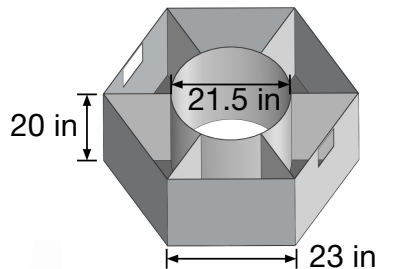


Proposed Mission Architecture

Spacecraft Structural Design

Design Overview:

- Hexagonal structure for spin stabilization
- Load bearing central thrust tube
- Articulating magnetometer boom
- 3mm Al 7075 wall panels



QwkSep 24
payload
separation
mechanism (300
kg mass limit)

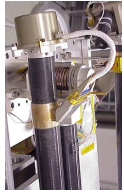


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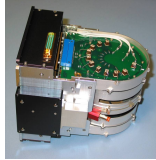
Payload Instrumentation & Onboard Computing

Instrument Suite Design



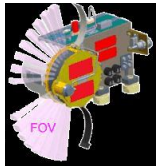
FGM

Fluxgate Magnetometer



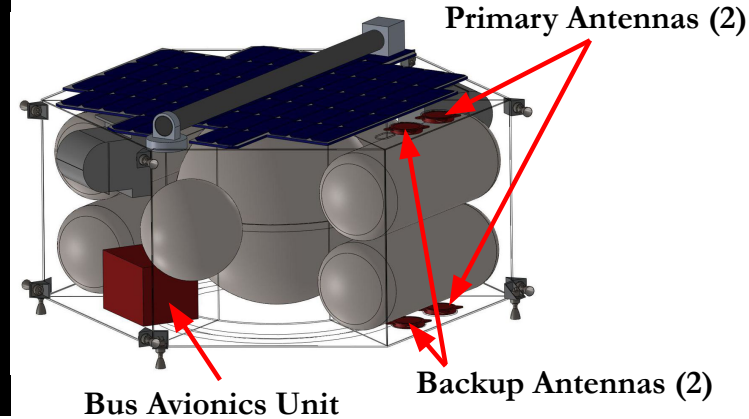
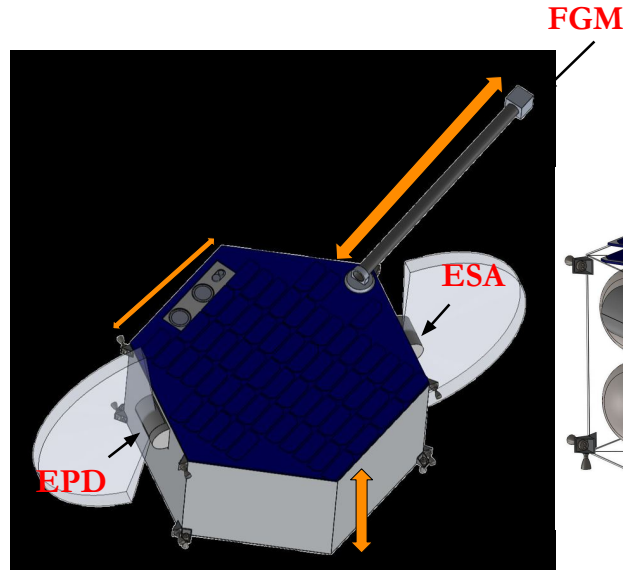
ESA

Electrostatic Analyzer

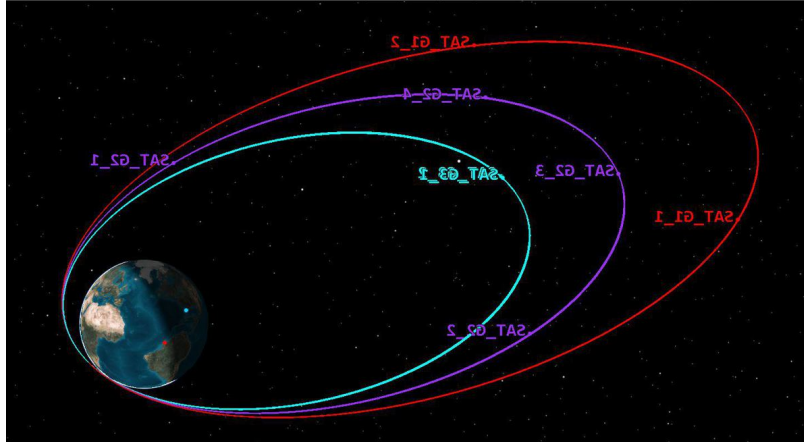


EPD

Energetic Particle Detector



Orbit Design

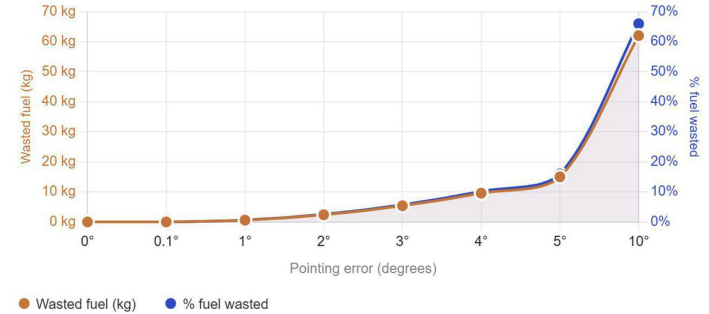


*Nominal
Orbits
Visualized
via
STK*

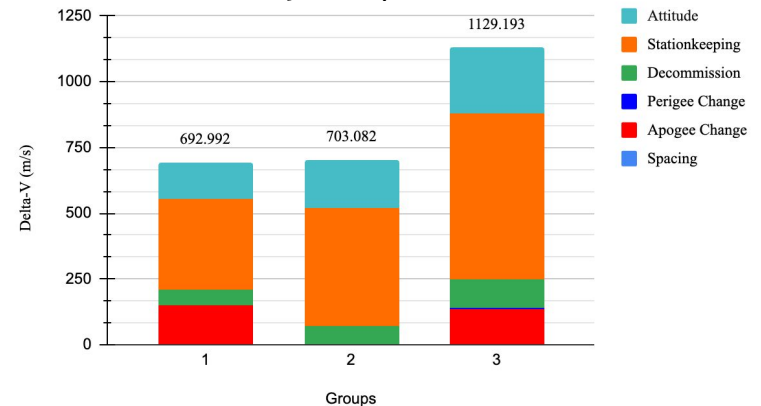
Group	R_a (km)	R_a (R_E)	R_p (km)	R_p (R_E)	T (hrs)
1	77104.160	12.089	7378.0	1.157	24.0
2	57143.955	8.960	7328.0	1.149	16.0
3	45942.426	7.203	7278.0	1.141	12.0

Nominal Orbital Parameters

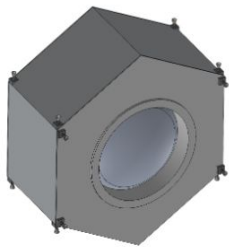
Pointing Accuracy



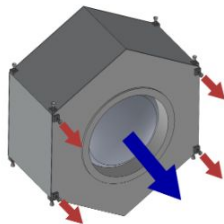
Delta-V Cost By Group



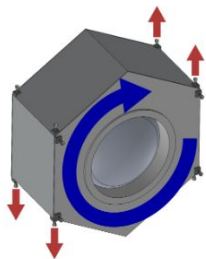
Spacecraft Attitude Determination



(a) Thruster Orientation

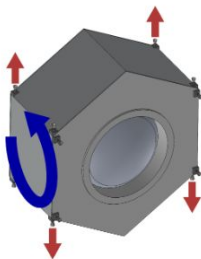


(b) Forward Flight



(c) Spin Up

Firing Maneuvers



(d) Spin Axis Slewing

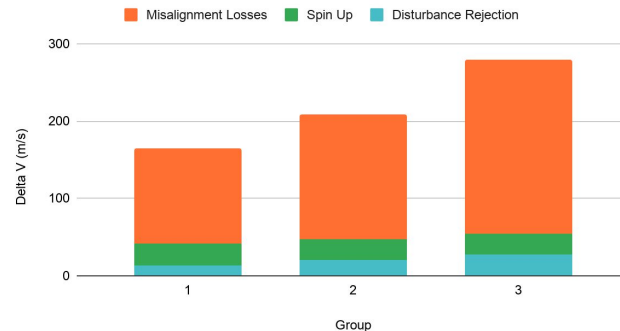


*Redwire Space
Fine Spinning
Sun Sensor*



DMU41

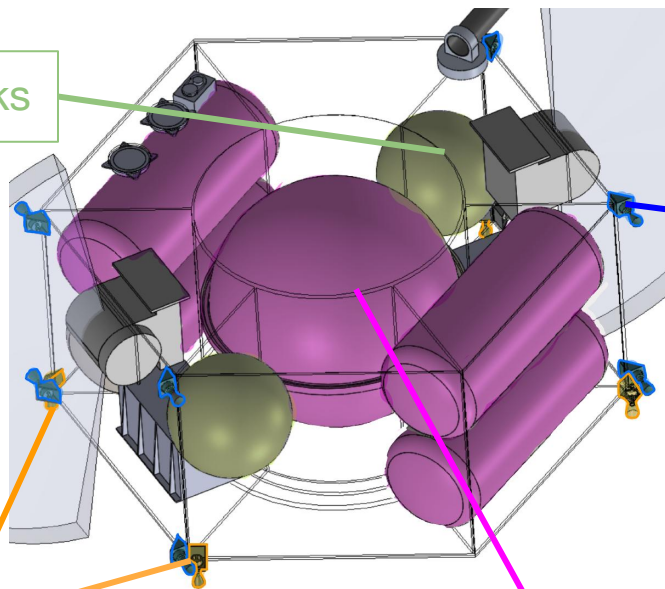
ADCS Delta V Budget



- Spin Stabilized @ 15 rpm
- 4 radial 1N thruster
- 8 axial 100 mN thrusters
- 2 Sun Sensors
- 1 IMU
- Passive Nutation Damper
- MEKF + PD

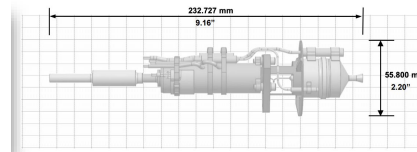
Propulsion & Launch Vehicle Design

Pressurant Tanks



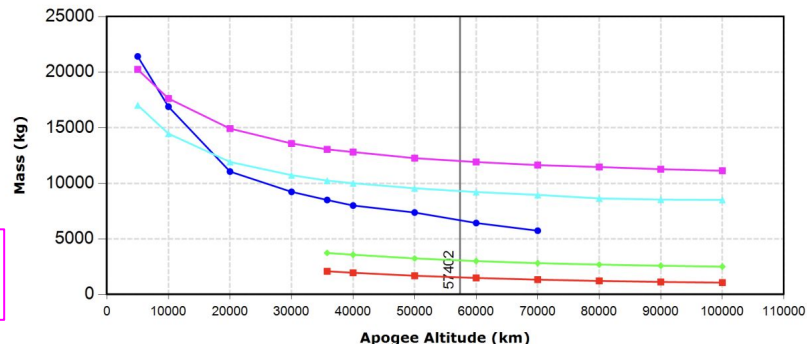
Moog Monarc-1 (1N) X 4

Hydrazine Tank (153 L)



L3Harris MR401 (0.1 N) X 8

Falcon 9 (Full Thrust, RTLS - 2000 km perigee) New Glenn (New Glenn - 2000 km perigee) Vulcan (VC4) Vulcan (VC6)
 Falcon 9 (Full Thrust, ASDS - 2000 km perigee)



Performance (kg) vs. Orbital Insertion Altitude (km) for NASA LSP Vehicles

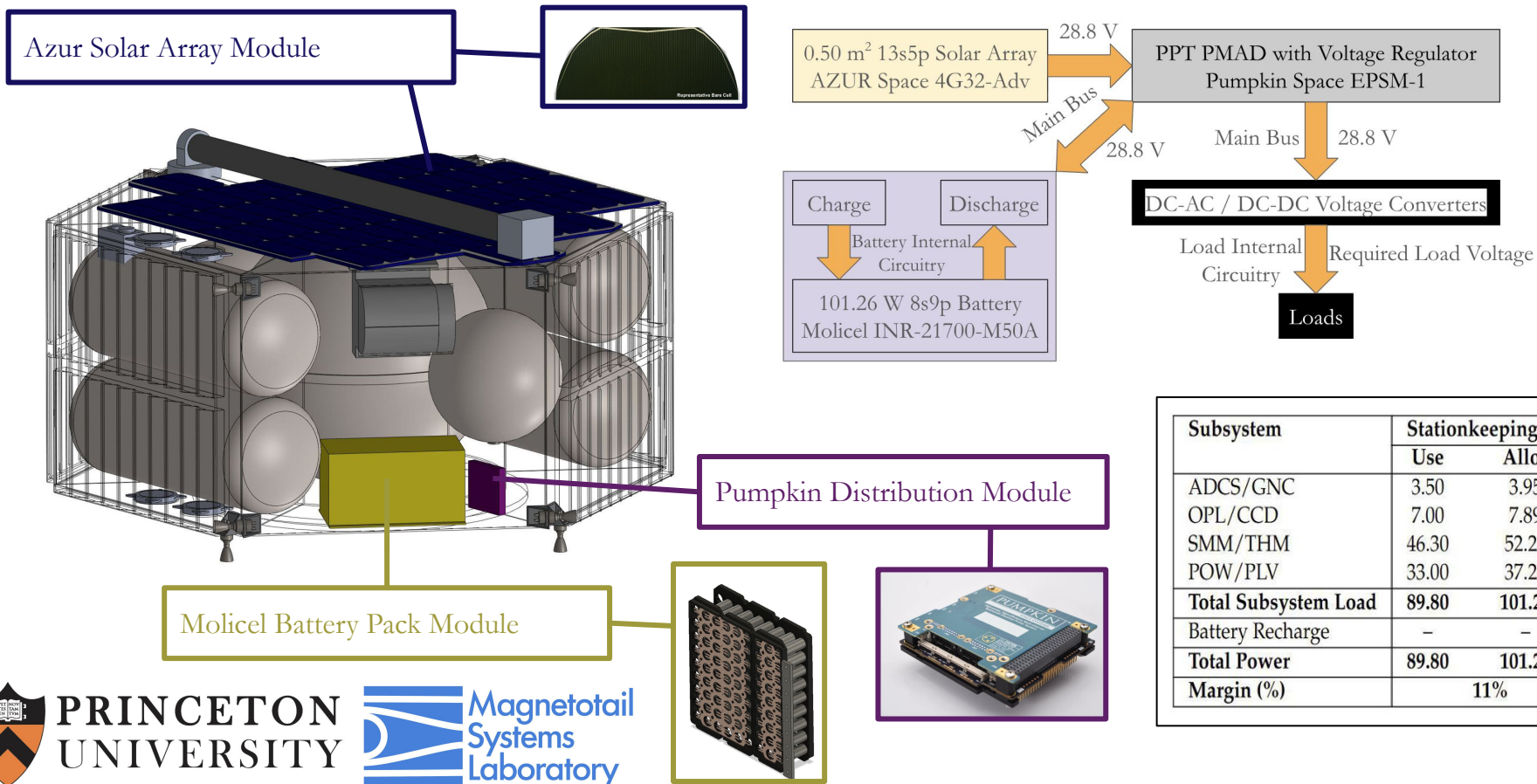
- Scenario 3 suggests NASA will select the Falcon 9 as the LV



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Power Design & Architecture Overview

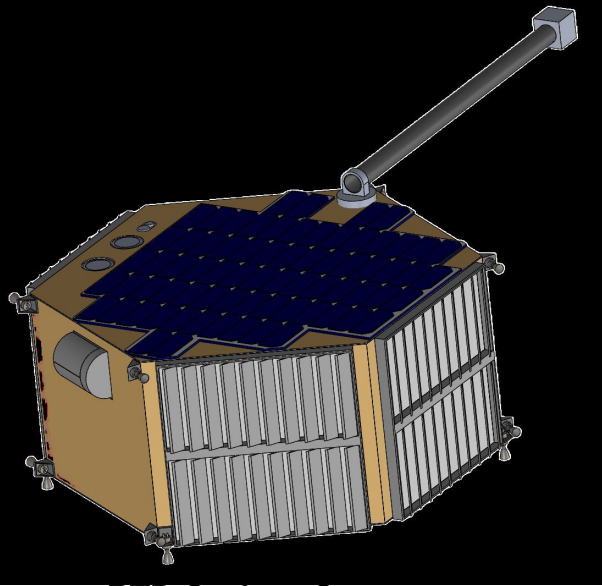


Thermal

Kapton MLI



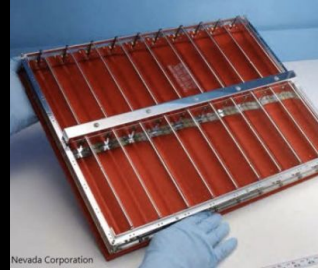
G-10



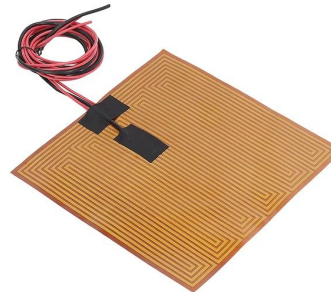
RTD Surface Sensors



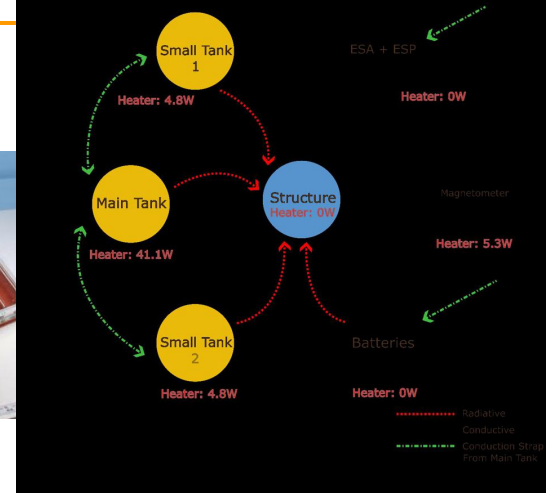
Louvers



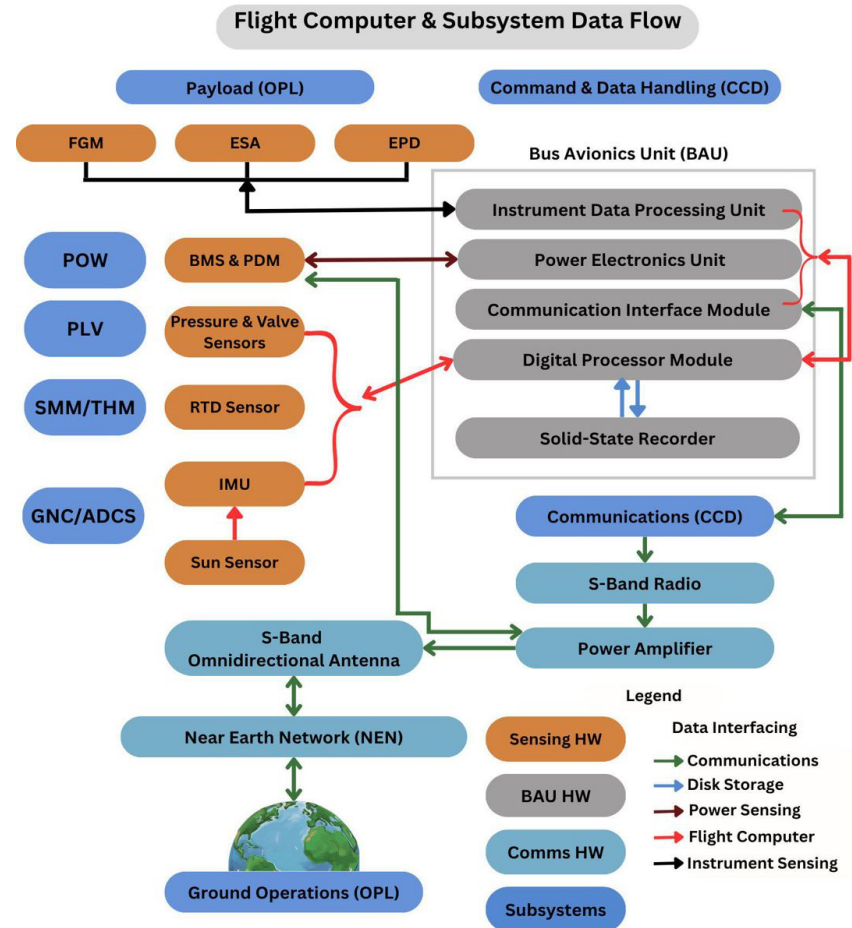
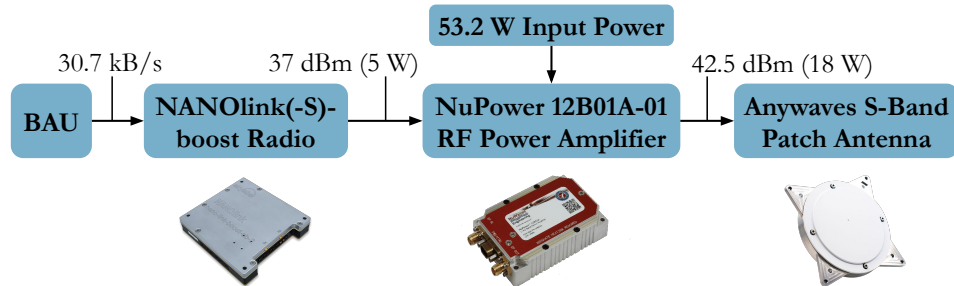
Thin Film Heaters



Graphite Thermal Straps



Flight Computer & Subsystem Interfacing



Mission Compliance

Requirements Compliance

In Compliance

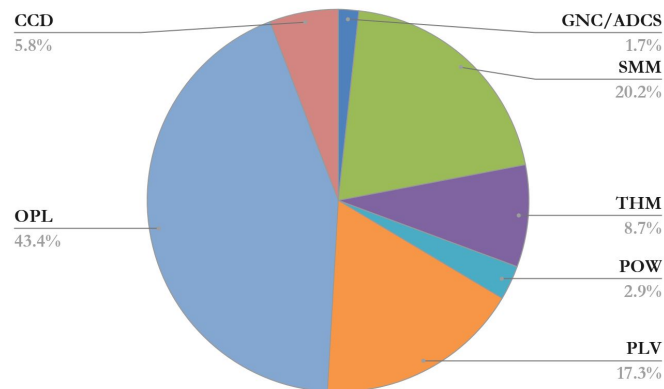
Requirement	Requirement ID	Compliance Demonstrated
Must obtain in-situ magnetic field and plasma measurements in the magnetotail transition region	MISS-F-001	• The three spacecraft groups (1, 2, 3) spend approximately 79.73%, 66.06%, and 47.82% of their time above $6 R_{\text{Earth}}$, allowing in-situ for measurements when their apogees are aligned with the magnetotail. • During this science window, the ESA, EPD, and FGM instruments are active.
Must make mesoscale spatial measurements ($1-3 R_{\text{Earth}}$) of the magnetotail transition region	MISS-P-002	• 8-spacecraft constellation across 7, 9, and 12 RE orbits enables multi-point sampling of the region • $1 R_{\text{Earth}}$ spacecraft spacing achieved through orbital phasing and ΔV maneuvers by ADCS and GNC
The MIDEX-class mission must be operational for at least 6 years	MISS-F-004	• Radiation-tolerant components (eg, ≥ 300 krad-rated electronics) exceed expected mission dose • Structural and propulsion analyses confirm mass, ΔV, and material durability sufficient for full mission duration



Budget Compliance: Cost

Subsystem	Budget (\$M)	Bill of Materials Total (\$M)	% Budget Used	Compliance	% Margin
GNC/ADCS	3	0.17	5.67	YES	94.33
SMM	35	24.02	68.62	YES	31.38
THM	15	0.8528	5.69	YES	94.31
POW	5	0.62426944	12.49	YES	87.51
PLV	30	15.24622872	50.82	YES	49.18
OPL	75	71	94.22	YES	5.78
CCD	10	3	29.91	YES	70.09

Budget Allocation (\$M)



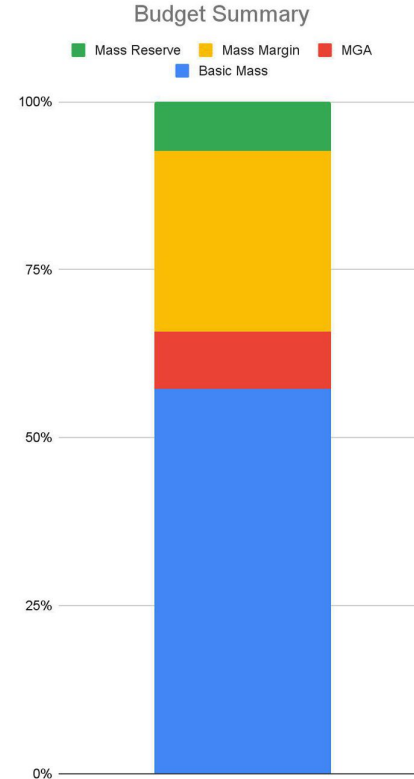
In Compliance

Budget Compliance: Mass

Subsystem	Mass (%)	Mass (kg)
CCD	1.0%	1.2
OPL	4.8%	5.7
GNC	0.0%	0.0
ADCS	4.5%	5.3
POW	7.8%	9.2
PLV	18.3%	21.7
SMM	55.5%	65.6
THM	8.0%	9.4
Total	100.00%	118.2

118.2 kg dry + 153 kg propellant = 271.2 total mass

In Compliance



Power Budget Compliance

Subsystem	Day (W)		Eclipse (W)		Transit (W)		Stationkeeping (W)		Safe Mode (W)	
	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc
ADCS/GNC	3.50	6.41	3.00	5.00	3.50	4.24	3.50	3.95	3.50	6.24
OPL/CCD	57.70	105.74	11.50	19.15	7.00	8.49	7.00	7.89	7.00	12.48
SMM/THM	1.00	1.83	46.30	77.11	1.00	1.21	46.30	52.21	46.30	82.54
POW/PLV	0.00	0.00	0.00	0.00	72.00	87.31	33.00	37.21	0.00	0.00
Total Subsystem Load	62.20	113.99	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Battery Recharge	29.61	29.61	–	–	–	–	–	–	–	–
Total Power	91.81	143.60	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Margin (%)	36%		40%		18%		11%		44%	

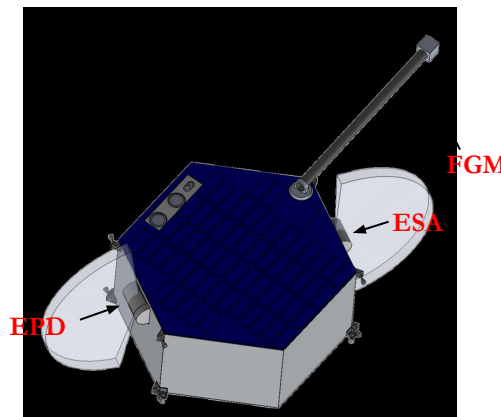
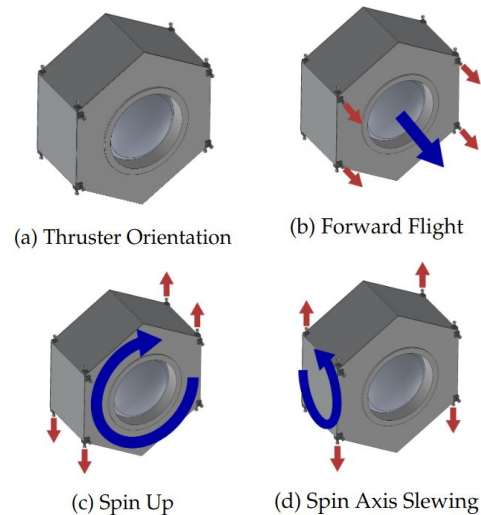
In Compliance



Pointing Compliance

In Compliance

- **Spacecraft is always sun pointing**
 - Maximizes solar power
 - Payload instruments view the magnetotail for ~50% of the year
 - Antenna gain is always >0 dB within $\pm 70^\circ$ pointing accuracy
- **Periodic stationkeeping burns (every 28 days)**
 - Spin down spacecraft
 - Receive telemetry update from Earth
 - Use attitude control thrusters to align thrust vector
 - Main engine burn
 - Re-establish sun-pointing
 - Spin up spacecraft



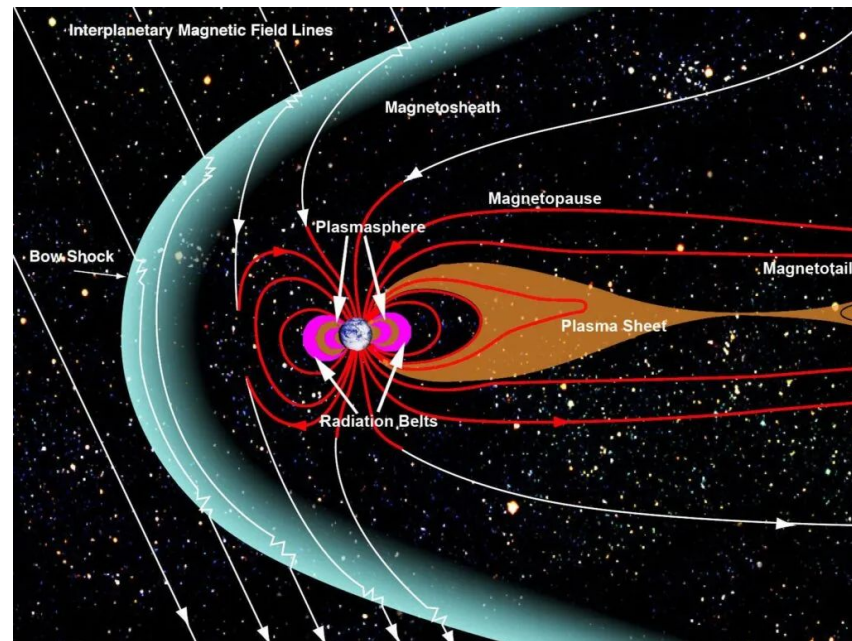
Concluding Remarks

Scientific Impact

- Provides the first coordinated, concurrent, multi-point mesoscale measurements of the 6–12 R_E magnetotail transition region resolving how BBFs evolve, brake, and deposit energy into the inner magnetosphere, thus fulfilling Topic D (Magnetotail Mesoscale).
- Also achieves Topic C (Magnetosphere Mesoscale) via orbit design and natural procession.

A Feasible, MIDEX-Class Architecture

- Eight spacecraft across three coplanar orbits (7, 9, and 12 R_E apogees) launched via NASA LSP, enabling simultaneous microscale, mesoscale, and upstream sampling without requiring station-keeping.
- All mission budgets are in compliance with positive margin.
- Design is mature and ready for Phase B detailed design and I&T, targeting a Feb 2033 launch and ≥ 5 -year science phase.



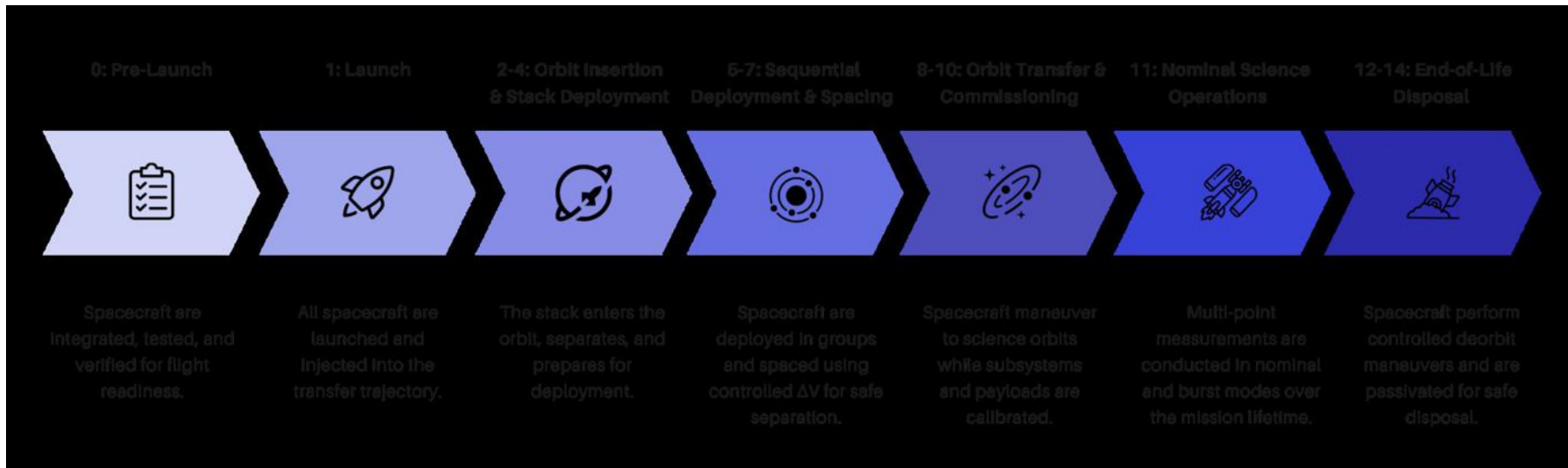
Earth's Magnetosphere and Plasmasheet, NASA [3]



Supplementary Slides

MATRIX Mission Operational Phases

The **key operational phases** are outlined in the diagram below. **14 total phases** span the course of 12 years between **2026 and 2038**



OPL/CCD: Definition of Spacecraft Modes & Procedures

Startup mode: *Deploy (D)+0 to D+12 h*

1. Power-on & boot · T+0 to T+30 s
2. Magnetorquer detumble to $< 2^\circ/\text{s}$ · T+30 s to T+2 h
3. Sun acquisition & 15 rpm spin-up · T+2 to T+4 h
4. First NEN contact, beacon on, IOD solution · T+4 to T+6 h → Commissioning

Commissioning mode: *T+6 h to T+9 months*

1. Bus health checkout (POW, THM, CCD, ADCS) · week 1
2. Magnetometer boom deploy & in-flight cleanliness cal · week 2
3. ESA + EPD power-up and calibration sweeps · weeks 3–6
4. Orbit raising/lowering & group spacing · months 2–9 → Nominal

Nominal science mode: *per 16.11 h orbit*

1. Science ON entering ROI ($> 6 R_E$): FGM 4 Hz, ESA + EPD full cadence
2. Burst trigger on $|\partial B_z / \partial t|$ threshold: FGM up to 128 Hz, prioritised storage
3. Exit ROI → survey cadence; prep downlink queue · ~30 min before perigee
4. Perigee contact (~1 h): CCSDS downlink, command uplink, ranging for OD
5. Post-perigee cruise out; eclipse survival if in shadow → back to step 1

Eclipse mode: *up to 4.356 h*

1. ESA + EPD off at shadow entry
2. FGM low-cadence survey only
3. Heaters on; battery to 30% DoD floor

Maneuver mode *per burn*

1. Payload off · thrust-vector pointing
2. Burn (1 N main or 100 mN ACS)
3. Re-acquire spin, resume nominal

Safe mode: *$t_{\text{response}} \leq 600\text{s}$*

1. Fault detected → payload off
2. Sun-point attitude re-acquired
3. Beacon + await ground uplink

End-of-life *after Dec 2038*

1. Apogee burn lowers perigee $< 300\text{ km}$
2. Passivate: vent propellant, drain batt.
3. Atmospheric re-entry & burnup



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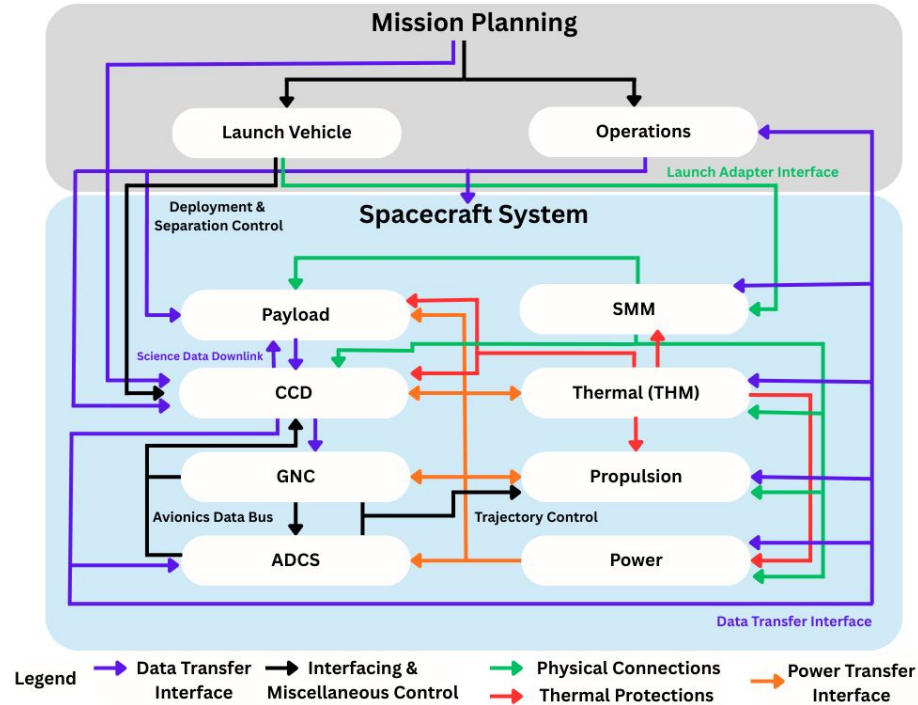


OPL/CCD: Pre-Launch Timeline

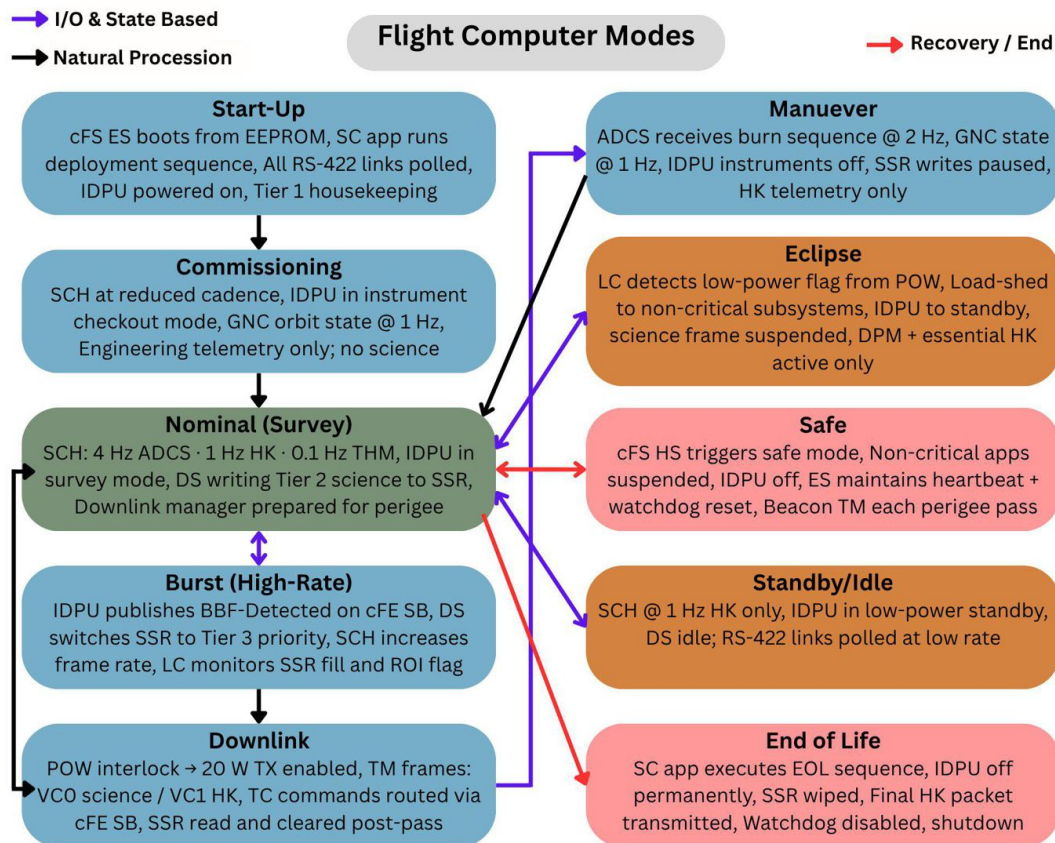


*Generative AI was used to create this diagram.

Systems Integration



OPL/CCD: Definition of Flight Computer Modes & Procedures



OPL/CCD: Payload & CCD Configurations

Payload Modes & Configuration

	ESA	EPD	FGM
Start-Up Mode	OFF	OFF	OFF
Commissioning Mode	OFF	OFF	Low Cadence
Nominal (Survey)	ON	ON	Low Field Range, LC
Downlink Mode	ON	ON	High Field Range
Eclipse Mode	ON	ON	LC/HC, Nominal/Burst
Maneuver Mode	OFF	OFF	OFF
Safe Mode	OFF	OFF	OFF
Standby/Idle Mode	OFF	OFF	Low Cadence

CCD Configuration

	IDPU	DCM	Radio/Antenna
Start-Up Mode	OFF	ON	OFF
Commissioning Mode	ON	ON	Low Duty TX/RX
Nominal (Survey)	ON	ON	Periodic TX
Downlink Mode	ON	ON	Full TX
Burst Mode	ON	ON	Buffering
Eclipse Mode	ON	ON	Minimal HK only
Maneuver Mode	OFF	ON	RX only
Safe Mode	OFF	ON	Low Duty TX/RX
Standby/Idle Mode	OFF	ON	RX only



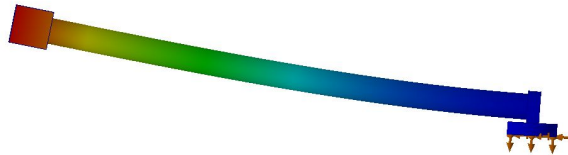
Additional Analyses and Results: SMM/THM

Hexagonal structure with trust tube and external

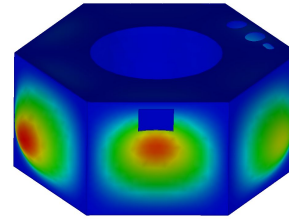
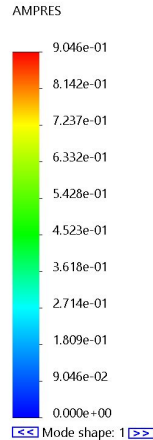
- Added cutouts for instrument/component FOVs
- Included massive/large volume components

FEA to determine areas of high stress

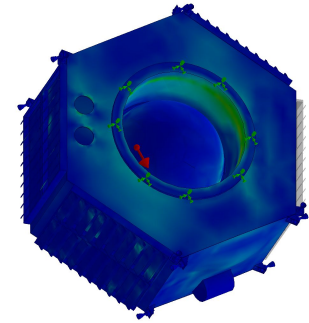
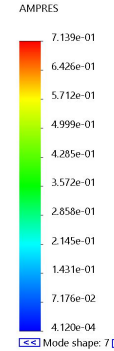
- Component brackets
- Wall panels



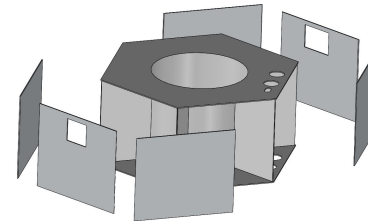
Natural frequency ~14 Hz



Unconstrained modal analysis - vibration of wall panels



Axial and lateral launch load static stress analysis

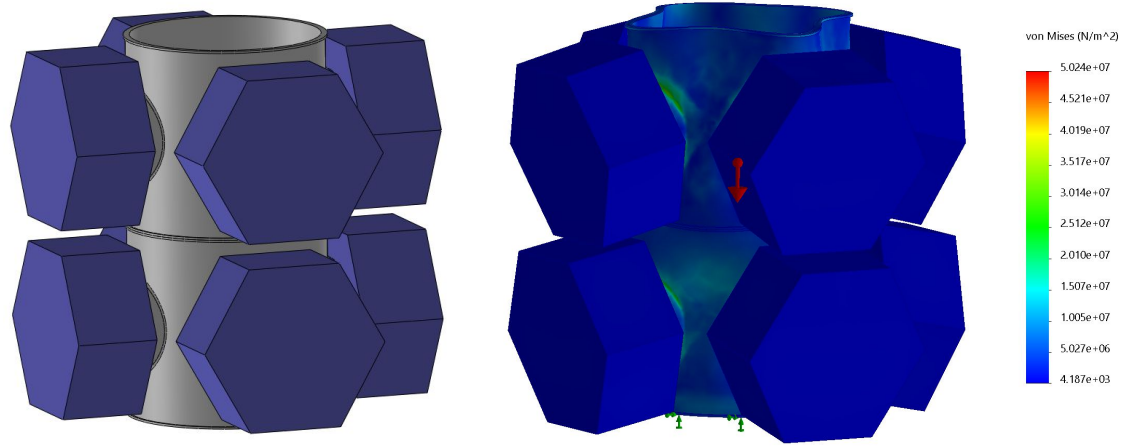


Additional Analyses and Results: SMM/THM

Launch Vehicle
Configuration

ESPA-derived payload
adapter

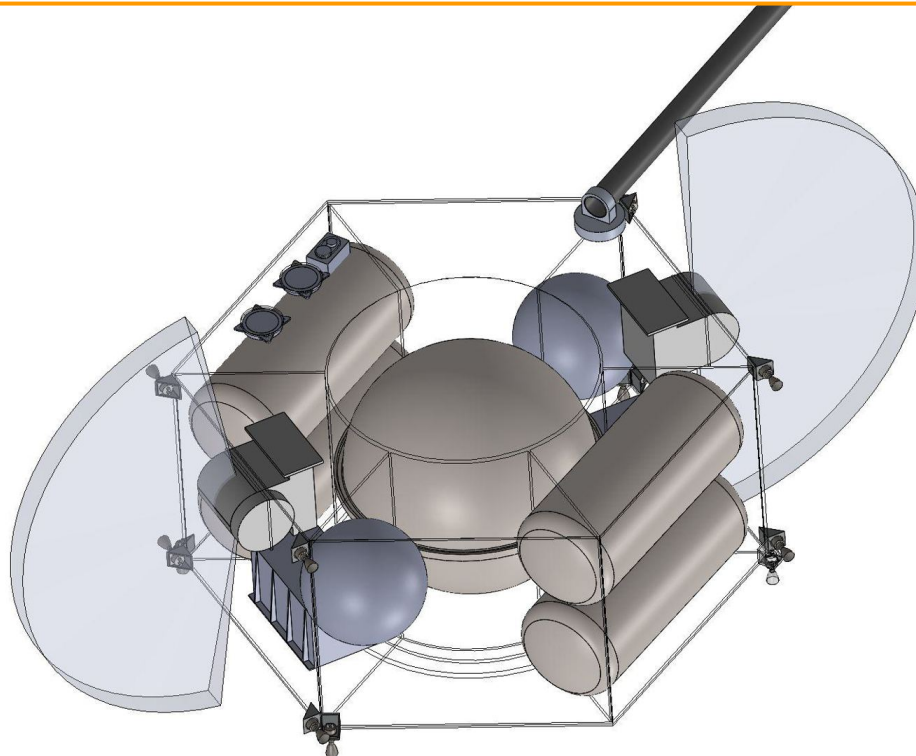
Aluminum 7075



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FOV and Component Layout

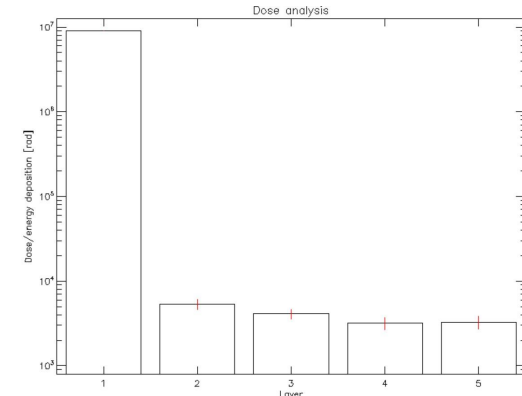
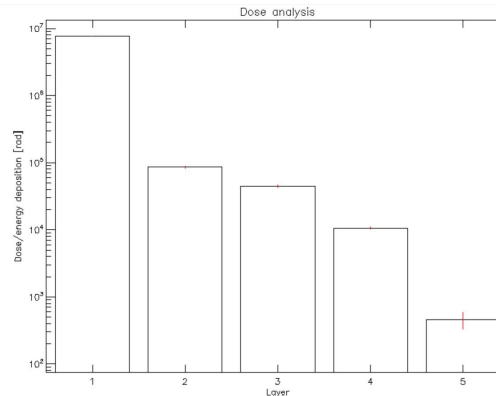
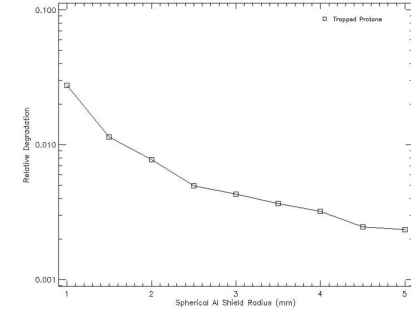


Refined Radiation Shielding Analysis

3 mm wall of Al7075 for radiation shielding + 2 mm thick AlTiTa enclosures around batteries and BAU unit → protection within allowed mass limits

Property	Al 7075 AL7075, [4]	Al 6061 [4], [5]	Lead [6], [7]	Fiberglass [8]	CF Composites [9], [10]	Polyethylene [11], [12], [13]
Density [g/cm ³]	2.81	2.70	11.35	2.11–2.60	1.15–2.25	0.93
Mass Atten. Coef. [cm ² /g] @ 1 MeV	1.19×10^{-2}	1.19×10^{-2}	5.21×10^{-2}	4.82×10^{-2}	2.21×10^{-2}	1.89×10^{-2}
Thermal Conductivity [W/mK]	130.00	130.00	35.00	1.30	0.22–400	0.20–0.51
Specific Heat Capacity [J/g°C]	0.96	0.96	0.128	0.81	1.0–1.2	1.90
CTE [μm/m°C]	25.20	25.20	29.00	5.00	−0.7–18	130–200
Cost (relative)	Moderate	Cheap	Cheap	Expensive	Expensive	Moderate

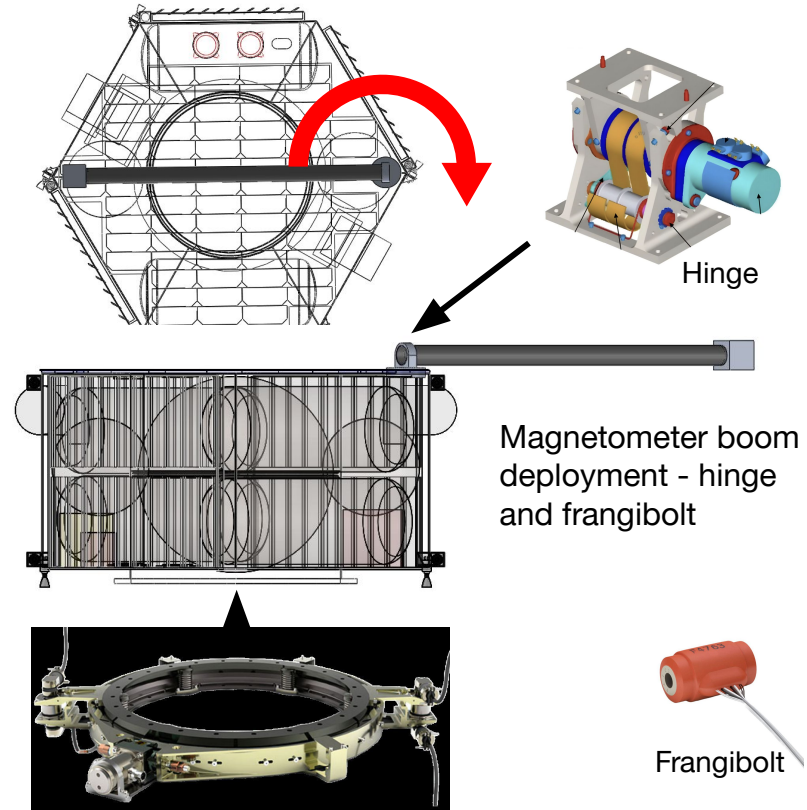
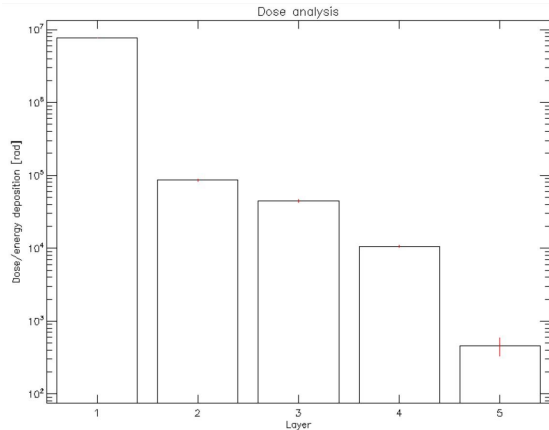
Table 3: Radiation Shielding Material Properties Comparison



Materials and Mechanisms

Material Selection: 3mm Al 7075 walls + 2mm AlTiTa radiation shielding around sensitive components (battery, BAU)

TID <<< 396 krad requirement



QwkSep 24 payload separation mechanism (300 kg mass limit)



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Material Selection Analysis

Property	Al 7075 [4]	Al 6061 [5]	Titanium [6]	Steel [7], [8]	Carbon Fiber Composites [9]
Density [g/cm ³]	2.81	2.70	4.43	7.86	1.15 – 2.25
Modulus of Elasticity [GPa]	71.7	68.9	114	196	2.62 – 520
Ultimate Tensile Strength [MPa]	572	310	900	160	0.917 – 4140
Tensile Yield Strength [MPa]	503	276	830	115	4.62 – 3380
CTE [°C ⁻¹]	2.36×10^{-5}	2.36×10^{-5}	8.6×10^{-6}	11.2×10^{-6}	$(-0.700 - 186) \times 10^{-6}$
Thermal Conductivity [W/mK]	130	167	6.7	17.9 – 22.6	0.220 – 400
Electrical Resistivity [Ω -cm]	5.15×10^{-6}	3.99×10^{-6}	1.78×10^{-4}	9.8×10^{-5}	$100 - 1.30 \times 10^{13}$
Relative Cost	Moderate	Low	Expensive	Expensive	Expensive

Table 3: Material properties

Al7075 chosen

lower density, greater ultimate tensile strength, and modulus of elasticity



Mechanisms

Launch Vehicle separation system analysis + Boom deployment mechanism

Parameter	QwkSep Pyrotechnic [22]	QwkSep Non-Pyro [22]	NEA Lightband System
Separation Shock	~1000 g	~100 g	~200 g
Operating Temperature Range	-68 to 104°C	-65 to 75°C	-40 to 85°C
System Mass	9.5 kg	9.5 kg	~12 kg
Payload Capacity	300 kg	300 kg	~400 kg
Actuation Type	NASA Standard Initiator	Mechanical Release	Non-Explosive Actuator
Integration Complexity	Moderate	Moderate	Higher

Table 6: Comparison of payload separation systems

Mechanism	Load Capacity	Shock	Reliability	Magnetic Cleanliness	Complexity	Thermal Sensitivity
Frangibolt	High	Moderate	High	Good	Moderate	Moderate
Pyrotechnic Cutter	Very High	High	Very High	Moderate	Moderate	Low
SMA Pin Puller	Moderate	Very Low	Moderate	Excellent	Low	High
Burn Wire	Low	Very Low	Moderate	Excellent	Very Low	High

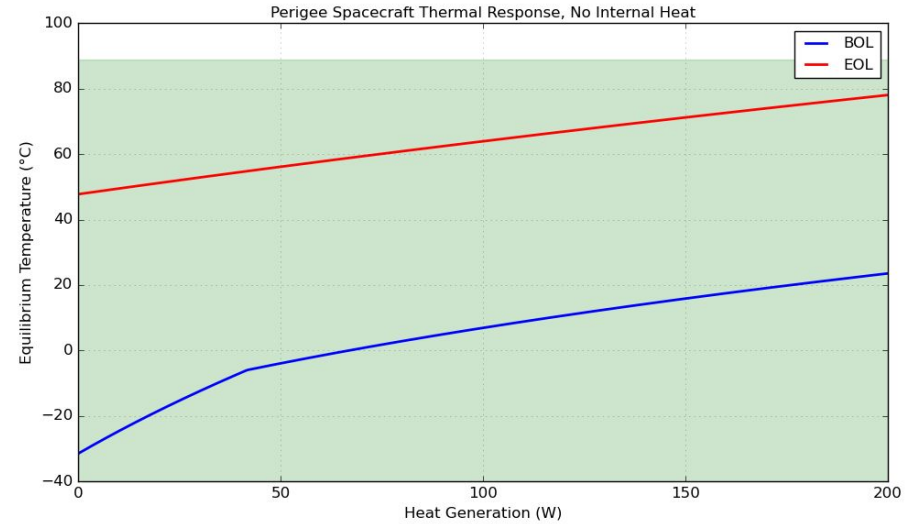
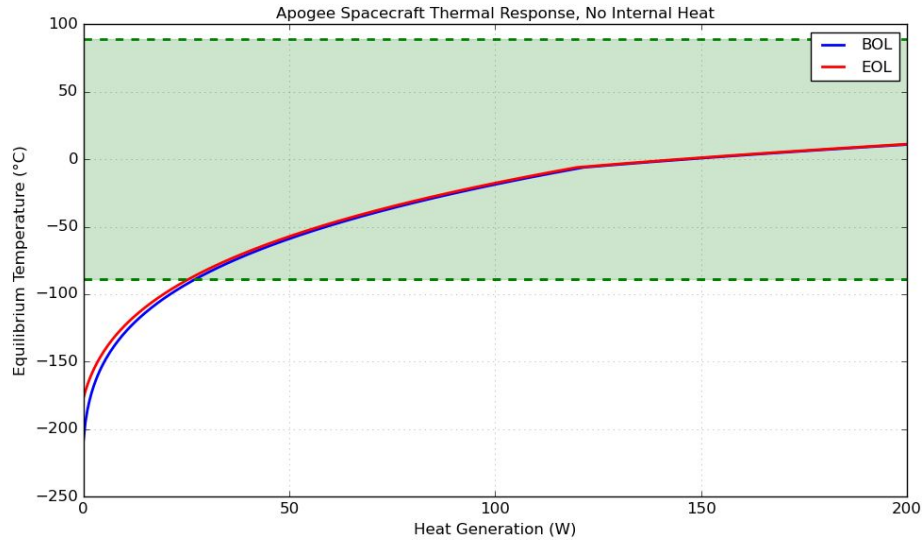
Table 3: Trade study comparison of magnetometer boom release mechanisms

Thermal Sensor Trade Study

Parameter	Thermistor (NTC) [13], [14]	RTD [15], [16]	Thermocouple [17], [18]
Temperature Range	-55 to +90 C	~ -125 to +170 C	~ -200 to +1250 C
Accuracy	High	High	Medium
Flight Heritage	Very High - See NASA GSFC S-311-P-18 part	High	Medium - more common in extreme scenarios (rocket nozzles or ground cryogenic)
Power Requirement	Requires small excitation current		No power required
Radiation Tolerance	~ 80 kGy	~ 300 kGy	~ 3000 kGy
Mass & Size	Varies depending on specific commercial sensors, but minimal mass and size		
Electrical Integration	Non-linear response	Linear response	Requires amplifier
Cost	~300/unit	~350/unit	~50/unit



Single-Node Model



MATRIX Optical Properties

System	Emissivity	Absorbtivity	
		BOL	EOL
Solar Cells	0.89	0.112	0.322
Magnetometer Boom	0.85	0.9	0.945
Thrusters	Not Used	Not Used	Not Used
Antenna	0.88	0.2	0.62
Payloads	0.85	0.9	0.945
ADCS Sensors	0.85	0.9	0.945
Louvers	0.14–0.85	Irrelevant	Irrelevant
MLI	0.01	0.04	0.31



Misc. Thermal Trade Studies

Technique	Solar Absorptivity (α)	IR Emissivity (ϵ)	α/ϵ Ratio	Thermal Conductivity
OSR (Optical Solar Reflector)	0.07–0.10 (BOL) 0.16–0.25 (EOL)	0.78–0.85	0.09–0.20	$\sim 1.0 \text{ W/m}\cdot\text{K}$ (borosilicate glass substrate) — not used for conduction
White Paint (e.g., S13G)	0.20 (BOL) 0.40 (EOL, 10k ESH)	0.9	0.22 (BOL) 0.44 (EOL)	$\sim 0.2\text{--}0.5 \text{ W/m}\cdot\text{K}$ (thin polymer paint film) — not used for conduction
Aluminized Kapton (tape/sheet)	$\sim 0.14\text{--}0.20$	$\sim 0.03\text{--}0.07$ (Al side)	$\sim 2\text{--}5$	$\sim 0.16 \text{ W/m}\cdot\text{K}$ (Kapton film, $25 \mu\text{m}$) — not used for conduction
MLI Blanket (20-layer typical)	$\sim 0.04\text{--}0.10$ (outer Kapton layer)	$\epsilon_{\text{eff}} \sim 0.01\text{--}0.03$ (system effective)	$\sim 2\text{--}5$ (outer layer)	
MLI with ITO Outer Layer	$\sim 0.10\text{--}0.15$	$\epsilon_{\text{eff}} \sim 0.01\text{--}0.03$	$\sim 3\text{--}7$	
Aluminum 6061 Film/Tape	0.11–0.12	0.45–0.73	0.24–0.16	$4.8 \text{ W/in}\cdot^\circ\text{C}$ ($\sim 188 \text{ W/m}\cdot\text{K}$)
Beryllium Heat Sink	0.31	0.03	10.3	$5.1 \text{ W/in}\cdot^\circ\text{C}$ ($\sim 200 \text{ W/m}\cdot\text{K}$)

Method	Cost	Mass Impact	Wattage	Passive?	Controllable?
Thin Film Heaters	Low	Low	3 W/in^2	No	Yes
Cartridge Heaters	High (Custom)	High	55 W/in^2	No	Yes
Conduction Straps	Moderate (Custom)	Moderate	Temperature Dependent	Yes	No

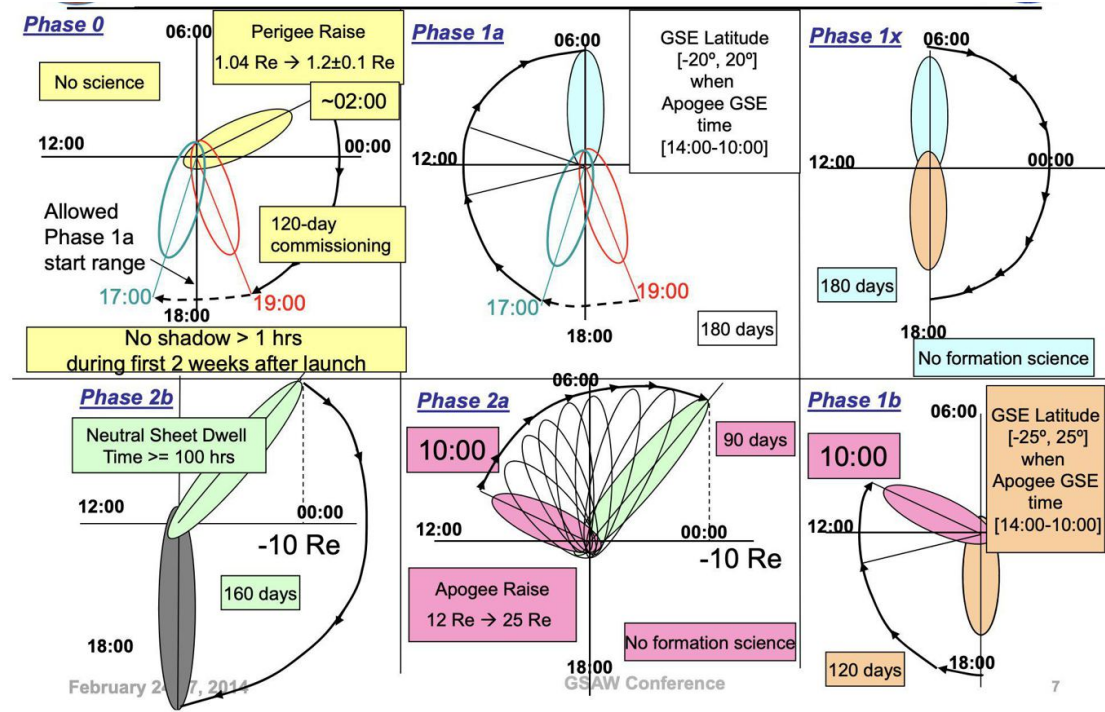


Design and Analysis - Satellite Spacing Temp Orbit

Temp Apogee (km)	Δr_a (km)	ΔT (sec)	Total Time for Separation (months)	Δv per Satellite (m/s)
57373.519	229.564	320.000	3	4.633
57313.919	169.965	240.000	4	3.475
57278.146	134.192	192.000	5	2.780
57254.292	110.337	160.000	6	2.317
57237.251	93.296	137.143	7	1.986
57224.468	80.513	120.000	8	1.738

Table 6: Separation Orbits

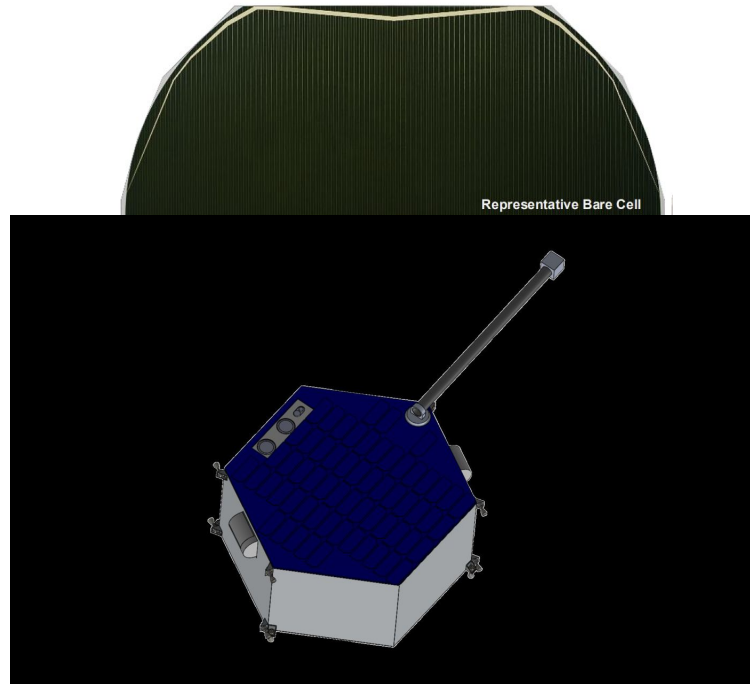
MMS ROI Comparison



Power Budget

Subsystem	Day (W)		Eclipse (W)		Transit (W)		Stationkeeping (W)		Safe Mode (W)	
	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc
ADCS/GNC	3.50	6.41	3.00	5.00	3.50	4.24	3.50	3.95	3.50	6.24
OPL/CCD	57.70	105.74	11.50	19.15	7.00	8.49	7.00	7.89	7.00	12.48
SMM/THM	1.00	1.83	46.30	77.11	1.00	1.21	46.30	52.21	46.30	82.54
POW/PLV	0.00	0.00	0.00	0.00	72.00	87.31	33.00	37.21	0.00	0.00
Total Subsystem Load	62.20	113.99	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Battery Recharge	29.61	29.61	–	–	–	–	–	–	–	–
Total Power	91.81	143.60	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Margin (%)	36%		40%		18%		11%		44%	

Power Generation - Solar Array Module



Group 3 (7 Re)

- AZUR Space 4G32-ADV solar cell
- 143.6 W total power generation
- 13 series cells
- 5 parallel cells
- 0.5 m^2 per spacecraft

Energy Storage - Battery Pack Module



Group 1 (12 Re)

- 4.356 hour eclipse
- Nominal voltage 28 V
- 8 series cells, 8 parallel cells
- Molicel INR-21700-M50A
- 4.48 kg
- 1661.60 cm³



Distribution & Management - Distribution Module



Distribution & Management
COTS PMAD, PPT Architecture
Pumpkin Space EPMSM-1



References

- [1] Bridgman, T., & Rastaetter, L. (2015, March 11). Magnetospheric Reconnection - July 2012. NASA Scientific Visualization Studio. <https://svs.gsfc.nasa.gov/4279/>. (cit. on slide 4)
- [2] Merkin, V. G., Panov, E. V., Sorathia, K. A., & Ukhorskiy, A. Y. (2019). Contribution of bursty bulk flows to the global dipolarization of the magnetotail during an isolated substorm. Journal of Geophysical Research Space Physics, 124(11), 8647–8668. <https://doi.org/10.1029/2019ja026872>. (cit. on slide 5)
- [3] Interrante, A., (2025, March 24). Earth's magnetosphere and plasmasheet. NASA Science. <https://science.nasa.gov/blogs/the-sun-spot/2023/09/26/earths-magnetosphere-and-plasmasheet/>. (cit. on slide 31)
- [4] Generative AI. Claude. Generation of Diagram. (cit. on slide 36)

Contributions

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Joseph Norwood: 17-18, 28

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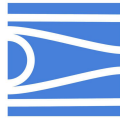
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